



IILM UNIVERSITY, GREATER NOIDA

Department of Computer Science & Engineering

INTERNSHIP REPORT

on

“Development of NOVA: A Responsive Registration Form with Client-Side Validation”

Submitted in partial fulfilment of the requirements for the
Degree of Bachelor of Technology
in Computer Science & Engineering

Submitted by

Vibhav Mishra

Roll No. 25SCS1003001169

B.Tech Computer Science & Engineering — Section 2CSE10

*Internship Organization: Codec Technologies Pvt. Ltd.
14 July 2026 – 14 August 2026*

DECLARATION

I, Vibhav Mishra, Roll No. 25SCS1003001169, a student of B.Tech Computer Science & Engineering, Section 2CSE10, IILM UNIVERSITY, GREATER NOIDA, hereby declare that the internship report titled “Development of NOVA: A Responsive Registration Form with Client-Side Validation” is an authentic record of the work carried out by me during my one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., from 14 July 2026 to 14 August 2026.

The information, project work and results presented in this report are based on my genuine effort and practical exposure gained during the internship period. This report has been prepared solely for academic purposes and has not been submitted elsewhere for the award of any other degree or diploma.

Place: Greater Noida
Date: 14 August 2026

Vibhav Mishra
Roll No. 25SCS1003001169
B.Tech CSE, Section 2CSE10

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to Codec Technologies Pvt. Ltd. for providing me the opportunity to undergo a one-month internship as a Web Developer Intern and for the practical exposure and guidance extended to me throughout the internship period, from 14 July 2026 to 14 August 2026.

I am thankful to the assigned project head(s) at Codec Technologies for their consistent support and valuable feedback, which helped me understand real-world frontend development practices and successfully complete my project, NOVA – a responsive registration form with client-side validation.

I would also like to express my heartfelt gratitude to the Department of Computer Science & Engineering, IILM UNIVERSITY, GREATER NOIDA, for providing the academic foundation and encouragement that made this internship possible. I sincerely thank the faculty members of the department for their continuous guidance and support throughout my B.Tech programme.

Finally, I extend my thanks to my family and friends for their constant encouragement, patience and motivation during the course of this internship and while preparing this report.

Vibhav Mishra

INTERNSHIP COMPLETION CERTIFICATE



INTERNSHIP OFFER LETTER



INTERNSHIP OFFER LETTER

Dear Vibhav Mishra,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Web Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 14/07/2026 to 14/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

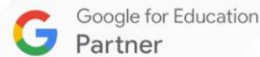
If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 14/07/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



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ABSTRACT

This report presents an account of the one-month internship undertaken at Codec Technologies Pvt. Ltd. as a Web Developer Intern, from 14 July 2026 to 14 August 2026, in Hybrid mode under the Web Development / Frontend Web Development domain. The internship provided practical, industry-oriented exposure to frontend web technologies, including HTML5, CSS3 and Vanilla JavaScript, and complemented the theoretical knowledge acquired during the B.Tech Computer Science & Engineering programme at IILM University, Greater Noida.

As part of the internship, a project titled “Development of NOVA: A Responsive Registration Form with Client-Side Validation” was undertaken. NOVA is a modern, responsive registration interface built using semantic HTML5 for structure, CSS3 for styling and responsive layout, and Vanilla JavaScript for client-side interactivity. The project incorporates real-time form validation, inline error feedback, password visibility toggling, a dynamic form-completion indicator, a success state upon valid submission, and a reset mechanism to register another account.

The report documents the weekly progression of work, the technical implementation of the NOVA interface, the testing carried out to verify validation behaviour, and the key technical and professional skills gained during the internship. The experience strengthened the author's practical understanding of frontend development, form usability and client-side validation techniques.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Internship

Industrial internships form an essential component of a B.Tech Computer Science & Engineering programme, bridging the gap between classroom learning and real-world professional practice. While academic coursework builds a strong theoretical foundation in programming, data structures and web technologies, an internship allows a student to apply that knowledge in a practical, outcome-driven setting. This experience helps students understand how individual concepts come together to produce a working, usable software product.

As part of this academic requirement, I undertook a one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., from 14 July 2026 to 14 August 2026, in Hybrid mode. The internship focused on Frontend Web Development, giving me hands-on exposure to building a complete, functional web interface from the ground up using HTML5, CSS3 and Vanilla JavaScript.

1.2 Objectives of the Internship

- To gain practical, hands-on experience in frontend web development.
- To apply classroom concepts of HTML, CSS and JavaScript to a real project.
- To understand the workflow of designing and building a responsive user interface.
- To learn and implement client-side form validation techniques.
- To develop discipline in testing, debugging and refining a web interface.
- To build confidence in independently completing a project within a fixed timeline.

1.3 Scope of the Report

This report documents the complete internship experience at Codec Technologies Pvt. Ltd., covering the organization profile, the week-by-week progression of work, and a detailed account of the project undertaken — NOVA, a responsive registration form with client-side validation. The report also discusses the tools and technologies used, the testing carried out, the learning outcomes achieved, the challenges faced, and the scope for future enhancement of the project. The report is limited to the frontend implementation of the NOVA interface and does not claim any backend, database or server-side functionality, as the current version of the project is strictly a frontend demonstration.

CHAPTER 2: ORGANIZATION PROFILE

2.1 About Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is the organization at which I completed my internship. As per the official offer letter issued to me, Codec Technologies is described as a platform dedicated to empowering learners and connecting talent with meaningful career opportunities, offering IT and business consultancy services. The internship program is affiliated with the National Internship Portal under the Ministry of Education and is recognized as an AICTE & ICAC approved internship program, as reflected on my Internship Certificate.

The internship offered structured, project-based learning in the Web Development domain, allowing interns to work on assigned projects and training tasks under the guidance of designated project heads, with performance reviewed on a weekly basis.

2.2 Internship Program Details

The table below summarizes the official details of my internship as confirmed by the Internship Offer Letter and the Internship Completion Certificate.

Particular	Details
Organization	Codec Technologies Pvt. Ltd.
Designation	Web Developer Intern
Duration	14 July 2026 – 14 August 2026
Duration Length	1 Month
Mode	Hybrid
Location	Pan India
Reporting To	Assigned Project Head(s)
Domain	Web Development / Frontend Web Development

CHAPTER 3: WEEKLY WORK DONE

The internship progressed in a structured, week-wise manner, moving from foundational orientation to interface development, JavaScript-based interactivity, and finally testing and finalization of the NOVA project. The activities described below reflect the practical, self-driven project work carried out during each week and are presented as a realistic beginner-level progression consistent with the Web Developer Intern role.

3.1 Week 1: Orientation & Foundation

Dates: 14 July 2026 – 20 July 2026

Main Focus: Getting oriented with the internship workflow and revising core frontend fundamentals.

Tasks / Work Performed:

- Attended internship orientation and understood the expected deliverables.
- Set up the development environment using Visual Studio Code.
- Revised HTML5 fundamentals and semantic webpage structure.
- Reviewed basic CSS3 concepts including selectors, box model and layout.
- Planned the structure and feature set of the NOVA registration project.

Technical Concepts Used:

- Semantic HTML5 tags
- Basic CSS3 syntax
- Project planning and requirement listing

Skills Gained: Familiarity with the development workflow and a clear plan for the NOVA project.

3.2 Week 2: Interface Development

Dates: 21 July 2026 – 27 July 2026

Main Focus: Building the structural and visual foundation of the NOVA registration interface.

Tasks / Work Performed:

- Built the HTML structure of the registration form, including all required input fields.
- Applied CSS3 styling for layout, spacing and typography.
- Implemented the dark-themed visual design and colour scheme of NOVA.
- Worked on responsive layout basics for different screen sizes.
- Refined the visual hierarchy of headings, labels and input fields.

Technical Concepts Used:

- HTML form elements (input, select, label)
- CSS Flexbox / layout techniques
- Responsive design basics
- Typography and colour theory

Skills Gained: Ability to translate a design concept into a structured, styled HTML/CSS interface.

3.3 Week 3: JavaScript & Validation

Dates: 28 July 2026 – 3 August 2026

Main Focus: Adding interactivity and client-side validation logic using Vanilla JavaScript.

Tasks / Work Performed:

- Implemented JavaScript for DOM interaction and event handling.
- Developed client-side validation rules for each form field.
- Added inline error messages for invalid or missing input.
- Implemented password visibility toggle (show/hide) functionality.
- Built a dynamic form-completion / progress indicator.
- Implemented the success state shown upon valid form submission.

Technical Concepts Used:

- DOM manipulation
- Event listeners
- Regular-expression-based validation
- Conditional UI rendering

Skills Gained: Practical understanding of client-side validation and interactive form behaviour.

3.4 Week 4: Testing & Finalization

Dates: 4 August 2026 – 14 August 2026

Main Focus: Testing the NOVA interface thoroughly and preparing final documentation.

Tasks / Work Performed:

- Conducted manual testing of all form fields under valid and invalid conditions.
- Tested responsive behaviour of the interface across screen sizes.
- Debugged validation issues identified during testing.
- Refined the UI based on testing observations.
- Prepared project documentation and internship report material.
- Prepared for project presentation and viva.

Technical Concepts Used:

- Manual functional testing
- Debugging techniques
- UI refinement
- Documentation practices

Skills Gained: Ability to systematically test, debug and finalize a frontend web project.

CHAPTER 4: PROJECT WORK

4.1 Project Title

Development of NOVA: A Responsive Registration Form with Client-Side Validation

4.2 Project Introduction

NOVA is a modern, responsive frontend registration interface developed during the internship using HTML5, CSS3 and Vanilla JavaScript. It is designed as a clean, structured and user-friendly registration experience, styled with a dark-themed, developer-onboarding aesthetic. The interface combines a semantic HTML5 structure with a visually engaging CSS3 layout and real-time client-side validation logic implemented in JavaScript.

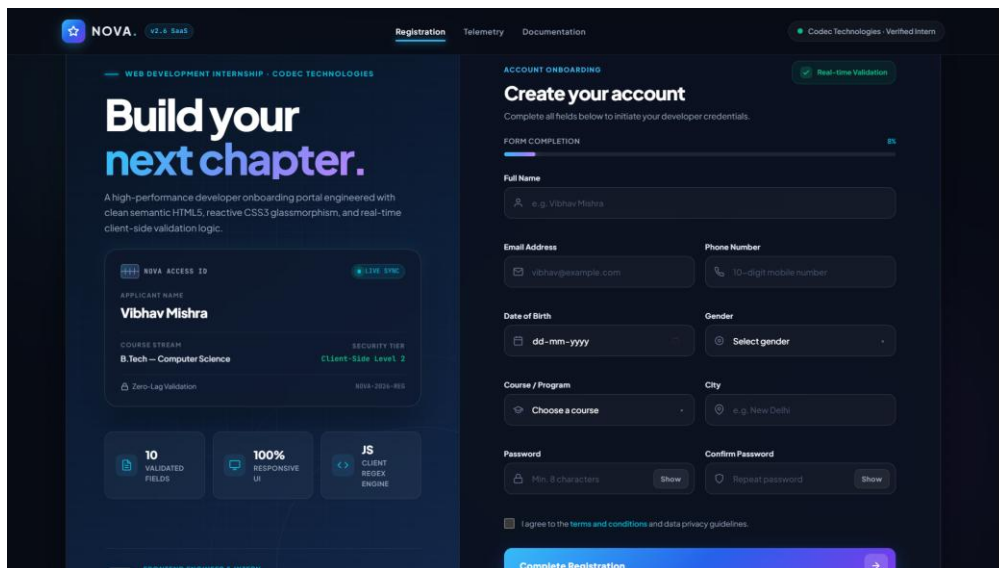


Figure 1: NOVA Registration Interface

4.3 Problem Statement

Many basic registration forms suffer from common usability issues such as poor visual organization, weak or absent validation, unclear feedback when a user makes an error, poor responsiveness across devices, and a lack of immediate guidance while filling out the form. Such shortcomings often lead to user frustration and incomplete or incorrect submissions.

NOVA addresses these issues at the frontend level by providing a clearly organized, visually structured layout, real-time client-side validation with inline error messages, a live form-completion indicator, and a responsive design that adapts to different screen sizes — together improving the overall usability of the registration process.

4.4 Objectives of the Project

- To design a modern, visually organized registration user interface.
- To implement a clearly structured and responsive form layout.
- To build robust client-side validation for all form fields.
- To provide immediate, field-level feedback to the user.
- To improve usability through interactive controls such as password visibility and a progress indicator.
- To apply HTML5, CSS3 and JavaScript concepts learned during the internship in a practical project.

4.5 Scope of the Project

Implemented in the current version:

- Frontend user interface for registration
- Responsive, structured form layout
- Client-side validation using JavaScript
- Interactive feedback (inline errors, password toggle, progress bar)
- Success state and reset / register-another-account functionality

Not implemented in the current version (discussed under Future Scope):

- Backend processing
- Database connectivity
- Real user authentication
- Server-side validation
- Persistent user storage

4.6 Features Implemented

The NOVA interface implements a range of features focused on collecting applicant information accurately and guiding the user through the registration process with clear, real-time feedback. The table below summarizes the key features implemented in the project.

Feature	Description
Full Name Field	Captures the applicant's full name with a minimum-length validation rule.
Email Address Field	Validates that the entered value follows a proper email format.
Phone Number Field	Accepts and validates a 10-digit mobile number.
Date of Birth Field	Captures the applicant's date of birth using a date picker.
Gender Selection	Dropdown selection for the applicant's gender.
Course / Program Selection	Dropdown selection for the applicant's academic course/program.
City Field	Captures the applicant's city of residence.
Password & Confirm Password	Enforces a minimum character length and validates that both fields match.
Password Show/Hide	Toggles password visibility using a "Show" control.
Terms & Conditions Checkbox	Requires explicit consent before registration can be completed.
Form Completion Indicator	Displays a live percentage bar reflecting how much of the form is complete.
Inline Validation Messages	Displays field-specific error messages immediately below each input.
Success State	Displays a confirmation screen with a summary of the validated details.
Reset / Register Another Account	Allows the user to reset the form and register a new entry.

Feature	Description
Responsive Layout	Interface adjusts to different screen sizes for a consistent experience.

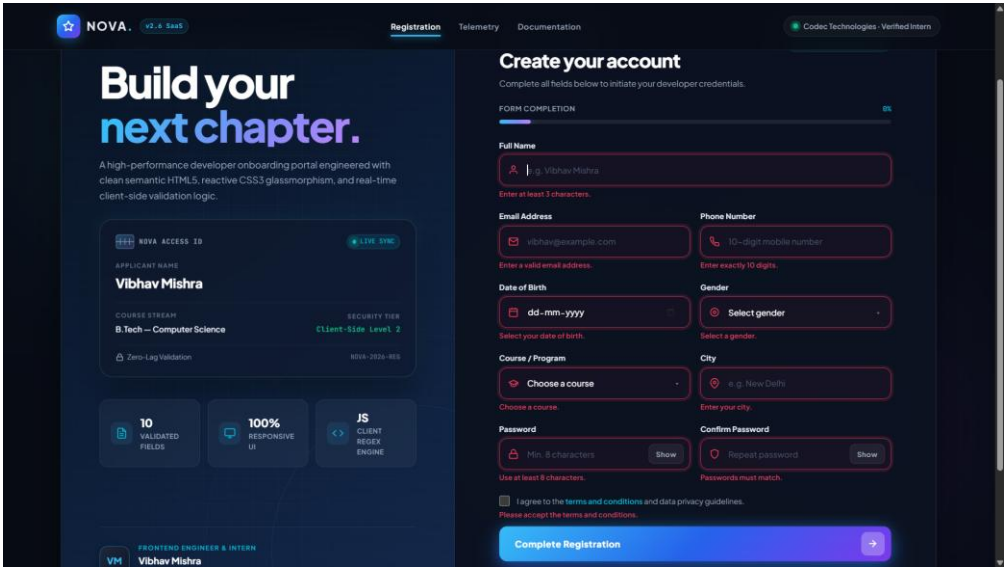


Figure 2: Client-Side Validation Error State

4.7 Functional Requirements

- Input Handling: The interface must accept and correctly capture user input across all form fields.
- Field Validation: Each field must be validated according to its specific rule (format, length, or required selection).
- Password Confirmation: The Confirm Password field must match the Password field exactly.
- Terms Acceptance: The user must explicitly accept the terms and conditions before submission.
- Error Feedback: Invalid or missing input must trigger a clear, field-specific inline error message.
- Successful Submission: Once all fields are valid, the form must display a success confirmation state.
- Reset Functionality: The user must be able to reset the form and register another account.

4.8 System Design / Architecture

As NOVA is a frontend-only project with no backend or database component, its architecture is intentionally simple and follows a linear flow from user interaction to validation and final state rendering, entirely within the browser.

USER → HTML5 INTERFACE → CSS3 PRESENTATION / RESPONSIVE LAYOUT → JAVASCRIPT LOGIC → VALIDATION & INTERACTION → ERROR STATE / SUCCESS STATE

4.9 Project Workflow

The registration workflow implemented in NOVA follows the sequence below, ensuring the user cannot proceed to a successful registration without passing all client-side validation checks.

Open NOVA Page → Enter Personal Information → Enter Account Details → Accept Terms & Conditions → Click Complete Registration → JavaScript Validates Input → (Invalid: Show Error Messages | Valid: Show Success State) → Register Another Account (Reset Form)

Figure 3: Validated Registration Form

4.10 Implementation Details

HTML5: Used to build the semantic structure of the page, including the registration form and its fields — full name, email, phone number, date of birth, gender, course, city, password and confirm password — along with the terms and conditions checkbox.

CSS3: Used for the overall dark-themed visual design, spacing, typography and layout of the form, as well as for making the interface responsive across different screen sizes.

JavaScript: Used to handle DOM interaction, implement validation rules for each field, display inline error messages, toggle password visibility, update the live form-completion indicator, and switch between the error state, the valid state and the success state.

4.11 Testing

Manual testing was carried out to verify that every field responds correctly to both valid and invalid input, and that the overall registration flow behaves as expected. The table below summarizes the key test cases executed on the NOVA interface.

Test Case	Input / Condition	Expected Result	Status
TC-01	Empty Full Name	Error message shown; field must be at least 3 characters	Pass
TC-02	Invalid Email format	“Enter a valid email address” error shown	Pass
TC-03	Invalid Phone Number	“Enter exactly 10 digits” error shown	Pass
TC-04	Missing Date of Birth	“Select your date of birth” error shown	Pass
TC-05	Missing Gender	“Select a gender” error shown	Pass
TC-06	Missing Course/Program	“Choose a course” error shown	Pass
TC-07	Missing City	“Enter your city” error shown	Pass
TC-08	Password below minimum length	“Use at least 8 characters” error shown	Pass

Test Case	Input / Condition	Expected Result	Status
TC-09	Confirm Password mismatch	“Passwords must match” error shown	Pass
TC-10	Terms & Conditions unchecked	“Please accept the terms and conditions” error shown	Pass
TC-11	All fields valid and complete	Form reaches 100% completion; success screen displayed	Pass

4.12 Results / Output

The final NOVA frontend successfully provides a structured registration form with a modern, dark-themed interface, a responsive layout that adjusts across screen sizes, real-time client-side validation with inline feedback, password show/hide functionality, a live form-completion indicator, a clear success state upon valid submission, and the ability to reset the form to register another account. As this is a frontend-only implementation, the project does not claim any backend or database storage of the submitted data.

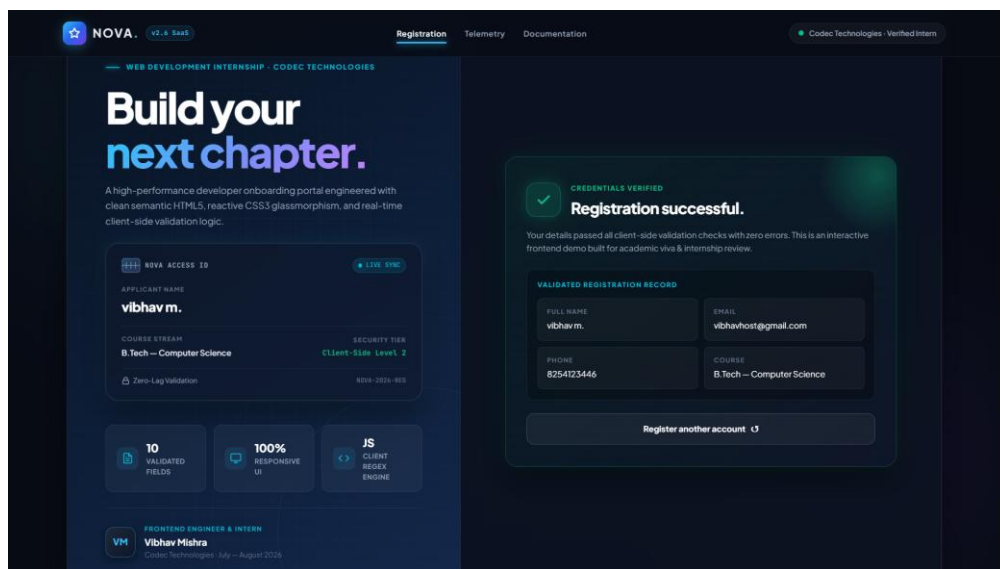


Figure 4: Successful Registration State

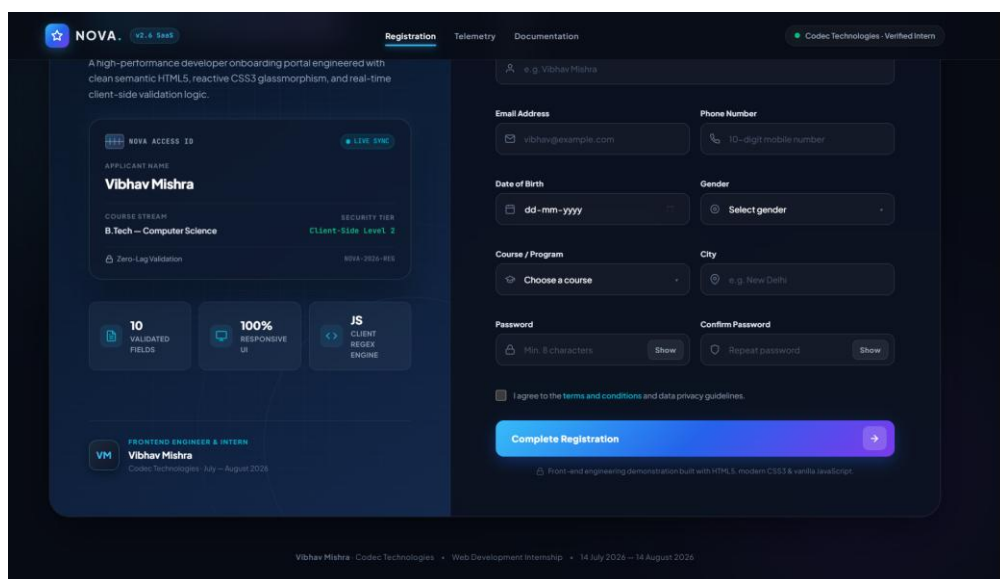


Figure 5: NOVA Responsive Frontend Interface

CHAPTER 5: TOOLS & TECHNOLOGIES USED

Category	Tool / Technology
Markup	HTML5 — semantic page structure and form fields
Styling	CSS3 — layout, responsive design, typography and visual styling
Scripting	Vanilla JavaScript — DOM manipulation and client-side validation
Development Environment	Visual Studio Code

Only the technologies listed above, along with the standard development environment used during coding, were applied in the NOVA project. No backend frameworks, database systems, or additional libraries were used, as the current implementation is strictly a frontend demonstration.

CHAPTER 6: LEARNING OUTCOMES

6.1 Technical Skills

- Structuring webpages using semantic HTML5.
- Styling and laying out interfaces using CSS3, including responsive design.
- Designing and structuring multi-field registration forms.
- Writing Vanilla JavaScript for DOM manipulation and event handling.
- Implementing client-side validation logic for various input types.
- Applying UI/UX fundamentals to improve usability and clarity.
- Debugging and testing a frontend interface systematically.
- Organizing and documenting a project for academic and professional presentation.

6.2 Professional / Soft Skills

- Time management while completing project milestones within a fixed one-month duration.
- Understanding and interpreting project requirements independently.
- Problem-solving while resolving validation and layout issues.
- Planning tasks in a structured, week-wise manner.
- Preparing clear technical documentation.
- Presenting completed work in a structured, professional format.

CHAPTER 7: CHALLENGES FACED

- Designing a clean, well-organized form layout for a registration interface with multiple fields.
- Maintaining consistent spacing and alignment across all form elements.
- Ensuring responsive behaviour of the interface across different screen sizes.
- Handling multiple validation rules simultaneously without conflicting error states.
- Writing clear, field-specific error messages that are genuinely helpful to the user.
- Managing JavaScript event handling for several interactive elements on a single form.
- Testing the interface against a wide range of valid and invalid input scenarios.
- Refining the user interface based on observations made during testing.

CHAPTER 8: FUTURE SCOPE

The current version of NOVA is a frontend-only implementation. The following enhancements are identified as future improvements and are not part of the current, completed implementation:

- Integration with a backend server to process registration requests.
- Connecting the form to a database for persistent storage of user data.
- Implementing real user authentication and account management.
- Adding server-side validation as a secondary layer of security.
- Incorporating email verification during the registration process.
- Implementing secure password handling and hashing on the server side.
- Building a user dashboard for registered applicants.
- Deploying NOVA as a fully functional, hosted web application.

CHAPTER 9: CONCLUSION

The one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., undertaken from 14 July 2026 to 14 August 2026, provided valuable, hands-on experience in frontend web development. Working in the Web Development / Frontend Web Development domain allowed me to move beyond theoretical classroom learning and apply HTML5, CSS3 and Vanilla JavaScript concepts to build a complete, functioning interface.

Through the development of NOVA — a responsive registration form with client-side validation — I gained practical experience in structuring web pages, designing responsive layouts, and implementing real-time validation logic using JavaScript. The process of testing and debugging the interface further strengthened my understanding of how a frontend application should behave under both valid and invalid user input.

Overall, this internship helped bridge the gap between academic learning and practical development, reinforcing my understanding of core web technologies while also improving my problem-solving, time-management and documentation skills. The experience has given me a stronger foundation to build upon as I continue to explore web development further in my academic and professional journey.

REFERENCES

Primary Sources

- Codec Technologies Pvt. Ltd. — Internship Offer Letter, issued 14 July 2026.
- Codec Technologies Pvt. Ltd. — Internship Completion Certificate, National Internship Portal, Ministry of Education.

Technical References

- MDN Web Docs — HTML, CSS and JavaScript Documentation, developer.mozilla.org
- W3C — Web Standards and Documentation, [w3.org](https://www.w3.org)